MULTIMODAL AI HACKATHON 2026 · TRACK B · MAPPING FLOOD DAMAGE FROM SPACE

FloodSight

Satellite-based flood, damage and connectivity assessment for rescue teams, demonstrated on the 26 August 2026 Trishuli (Bhote Koshi–Trishuli corridor) flood, Nepal.

Team	FUDICIA
Repository	https://github.com/Kishore Gauthaman-ai/Floodlight
Case study	Trishuli corridor, 84.39–85.73°E, 27.68–28.31°N
Frozen result state	`config b09870f7…bc1bb · source ba55a74a…3c00`
Checkpoint (v1, production)	`SHA-256 cb91fa63…13cbbd37`
License	MIT (code) · sources retain their own licenses

104 automated tests passing · **One-shot EMSR927 comparison, no post-hoc tuning** · **Every figure traced to a result file**

0.0 km² standing water — honest zero · **AI model not yet proven on mountainous terrain**

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> **How to read this report.** Every quantity in this document is produced by a script reading a result file written during the pipeline run — no number is estimated or recalled from memory. The full number→source mapping ships with the repository as `pdf_content_sources.txt`. Where a result is weak or unproven, this report states that plainly in the same section rather than only in the limitations section.

1. Summary

FloodSight answers three questions for any area and flood date: **where did the flood hit**, **what was probably damaged**, and **which settlements are cut off**. For the 26 August 2026 Trishuli flood, FloodSight ran its full pipeline end to end on real Sentinel-1 radar, the Copernicus GLO-30 DEM, and a pre-event OpenStreetMap snapshot.

STANDING FLOOD WATER	EVENT-ATTRIBUTED CHANGE	BUILDINGS HIT (HEADLINE)	BRIDGES FLAGGED	SETTLEMENTS POSSIBLY CUT OFF
0.0 km²	**0.455 km²**	**30**	**19**	**106**
True zero — water had receded	4.69× control (a), 3.07× control (d)	Centroid inside event layer; 82 touch it	Approach in zone — none observed damaged	0 at "likely" tier ≠ nobody affected

EMSR927 RECALL AT 100 M

0.34

Precision 0.76; see §7.3 for caveats

The post-event radar scene was acquired roughly two days after the flood; by then the flash flood had already receded, so FloodSight found **no standing flood water on this event** and does not claim otherwise. The radar pair instead shows surface change. A single before/after comparison flags **41.99 km²** of change, but control runs on non-event date pairs trigger a similar or larger area, so this layer is reported only as **"other ground change (monsoon baseline, not event-attributed)"** and is not treated as damage.

A second, independently validated method — a five-scene pre-event temporal stack with a threshold set purely from non-event controls — attributes **0.455 km²** of valley-floor change specifically to the event: 4.69× a pre-event control area (0.097 km²) and 3.07× an independent pre-event control (0.148 km²). This is the number used for the damage and cut-off analysis in this report.

Of 899 OpenStreetMap settlements inside the approved bounding box, 106 are possibly cut off, 111 are verified connected, 85 were already isolated before the flood, and 483 sit on routes crossing terrain the radar cannot reliably assess — they are reported as **unassessed**, not as connected. Zero settlements reach the stricter "likely cut off" confidence tier; this must not be read as "nobody is affected" (§6, §9).

The declared AI component, a U-Net trained on Kuro Siwo, scores IoU 0.63 on held-out test events but performs *worse than a simple radar-threshold baseline on the validation split*, and no held-out training or test event is mountainous — so Himalayan generalisation is demonstrated as an open question, not a solved one (§5). FloodSight is an educational prototype and is not an operational early-warning or damage-certification tool.

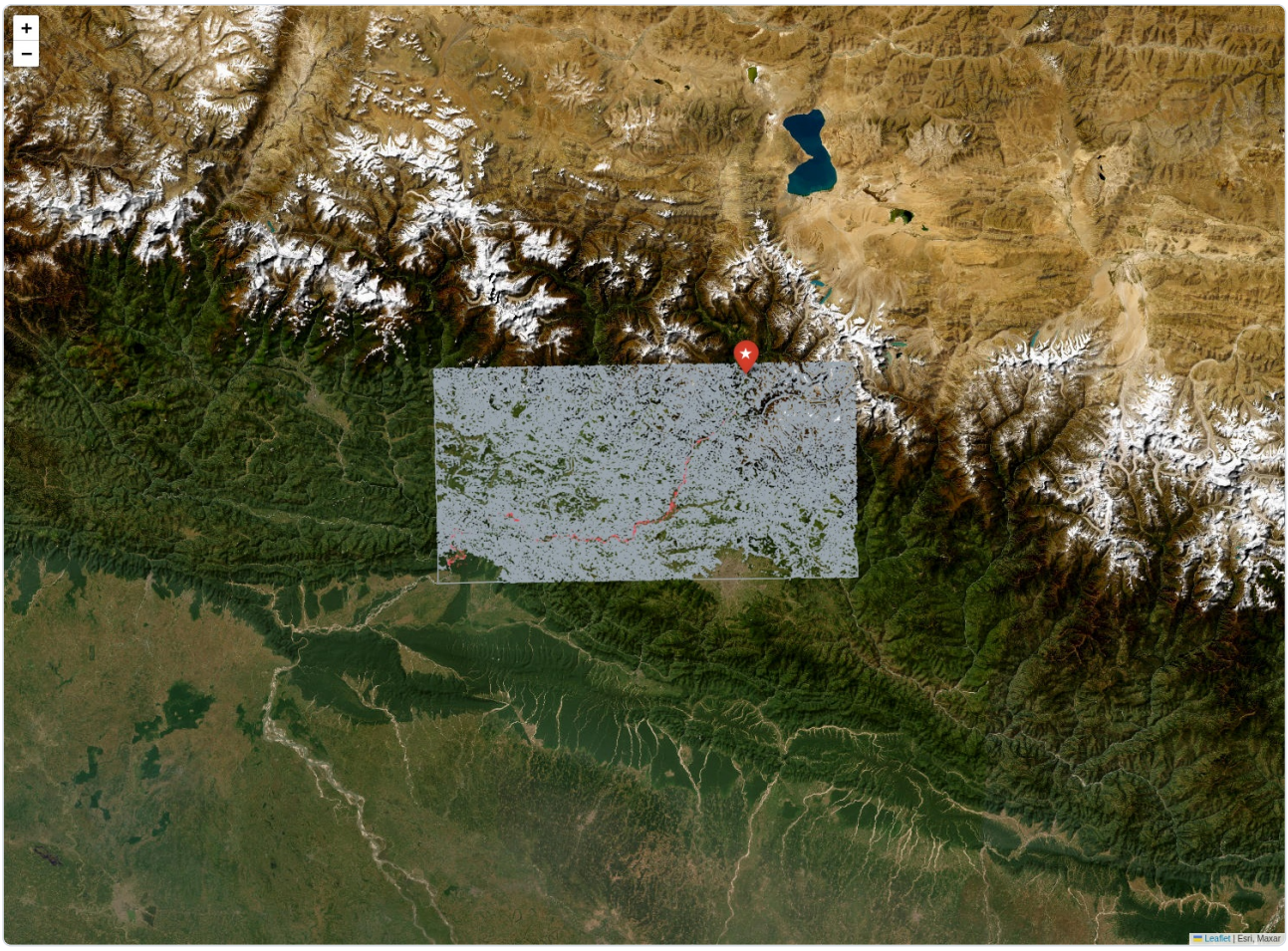


Figure 1. The approved Trishuli corridor bounding box (white outline), the radar-reliable analysed area (shaded), the event-attributed change layer (red), and the upstream point used for the flood-path trace (marker). Esri/Maxar satellite basemap.

2. Problem and approach

On 26 August 2026 a glacier collapse in northern Nepal sent a flood of water, mud and debris down the Bhote Koshi–Trishuli corridor, cutting roads, bridges and communications within hours. Rescue coordinators needed answers before ground survey teams could reach the area. Radar satellites can see through monsoon cloud cover; optical satellites usually cannot during the monsoon season itself.

Sentinel-2 optical scar detection was built and tested for this event, then deliberately stopped: joint cloud-free observability across the pre/post window was only **15.4%**, and the only pre-event scene usable for a baseline was 3% valid by area. No optical number is used anywhere in this report's case-study results; this was a measured decision, not an oversight (§8.2).

FloodSight's pipeline has seven layers:

1. **Input validation** — checks the bounding box, the flood date, and that the OpenStreetMap snapshot predates the event, before any download starts.
2. **Acquisition** — Sentinel-1 search with same-orbit-track pairing (cross-track comparisons are never made), Copernicus GLO-30 DEM tiles, and a dated pre-event OpenStreetMap extract.
3. **SNAP radar preprocessing** — orbit correction, thermal-noise removal, radiometric calibration, Lee speckle filtering, and 10 m terrain correction against GLO-30, run through ESA SNAP headless.
4. **Detection** — a trained U-Net, a classical log-ratio/Otsu baseline, and an event-attributed temporal-stack layer (§4).
5. **Damage and connectivity analysis** — overlay against the pre-event OpenStreetMap snapshot for buildings, roads and bridges, and a road-graph cut-off analysis for settlements.
6. **A schema-validated results store** — every number the dashboard, report and assistant show is read from this store; nothing is generated independently downstream.

7. **Presentation** — an interactive dashboard, bilingual (English/Nepali) one-page situation reports, and a template-grounded assistant.

Every stage fails visibly and caches by key; a failed or skipped stage is recorded with a reason rather than silently omitted.

3. Data and rule compliance

Allowed inputs used: Sentinel-1 and Sentinel-2 from the Copernicus Data Space Ecosystem, Copernicus GLO-30 DEM, a pre-event OpenStreetMap snapshot, and a documented Kuro Siwo training subset.

OpenStreetMap snapshot: the ohsome API returned persistent errors during development, so the pipeline uses a dated Geofabrik history extract instead, fixed to the state of **2026-07-01** and md5-verified. An automated check enforces that this date is strictly earlier than the flood date before any run proceeds.

EMSR927 isolation: the EMSR927 reference was read only during evaluation, after the pipeline's code and configuration were frozen, solely to check results that had already been produced. An automated test asserts that the pipeline source tree never imports from the evaluation/reference package. The same-orbit-track rule is enforced directly in the pairing code: a cross-track radar comparison cannot occur.

> **Compliance summary:** no OpenStreetMap edit made after the event enters the pipeline; no published damage map (EMSR927, UNOSAT, or any other) is used as a pipeline input at any stage; EMSR927 is used exactly once, for checking, after a code freeze.

4. Method

The pipeline is deterministic and resumable: every stage caches by its inputs (bounding box, date, scene IDs, code version). A failed stage stops the run visibly; stages that depend on it are skipped with a written reason rather than silently producing a partial result. No stage invents a value — unknown quantities are written as `null`, and the results store is validated against a schema on both write and read.

4.1 Radar preprocessing

ESA SNAP runs headless per scene: orbit correction, thermal-noise removal, calibration to sigma-nought, Lee speckle filtering, and 10 m terrain correction against the Copernicus GLO-30 DEM. Same-slice pairs are co-registered to a common grid; where a track required multiple slices to cover the analysis area, the slices are processed separately and unioned with per-slice attribution recorded in the provenance file.

In steep terrain, radar backscatter is unreliable: **43.85%** of the approved bounding box is masked for slope above 30°, with a further small fraction masked for layover and shadow. Every downstream statistic (debris area, damage counts, cut-off analysis) is clipped to the resulting **analysed area**, and nothing outside it is reported as "connected" or "not affected" — it is reported as **unassessed**.

4.2 Four detection streams

- **Trained U-Net** and **classical log-ratio/Otsu baseline** — standard flood-water detectors run on the single co-registered before/after pair (§5).
- **Single-pair change layer** (41.99 km² on this event) — any radar brightening near mapped waterways between the two scenes. This layer fires on every monsoon-season date pair tested, including dates with no flood, and is therefore relabelled **"other ground change (monsoon baseline, not event-attributed)"** rather than treated as damage.
- **Event-attributed layer** — five same-track pre-event scenes provide a per-pixel temporal median and median-absolute-deviation (MAD) of VV backscatter in dB. The post-event scene is converted to a per-pixel z-score against that pre-event distribution. The detection threshold (`z = 4.679`) is the 99.9th percentile of the *control* distribution — set from non-event data only, never from the EMSR927 reference. Attribution is further restricted to the valley floor: height above nearest drainage (HAND) ≤ 25 m, slope ≤ 15°, within 2 km of the Trishuli, Bhote Koshi or Langtang main stems.
- **Height-above-drainage (HAND) layer** — computed with a vectorised nearest-drainage-cell method and validated against a slower per-cell flow-walk reference implementation; the two agree to within **3.35 m** in the valley-floor regime that matters for this analysis.

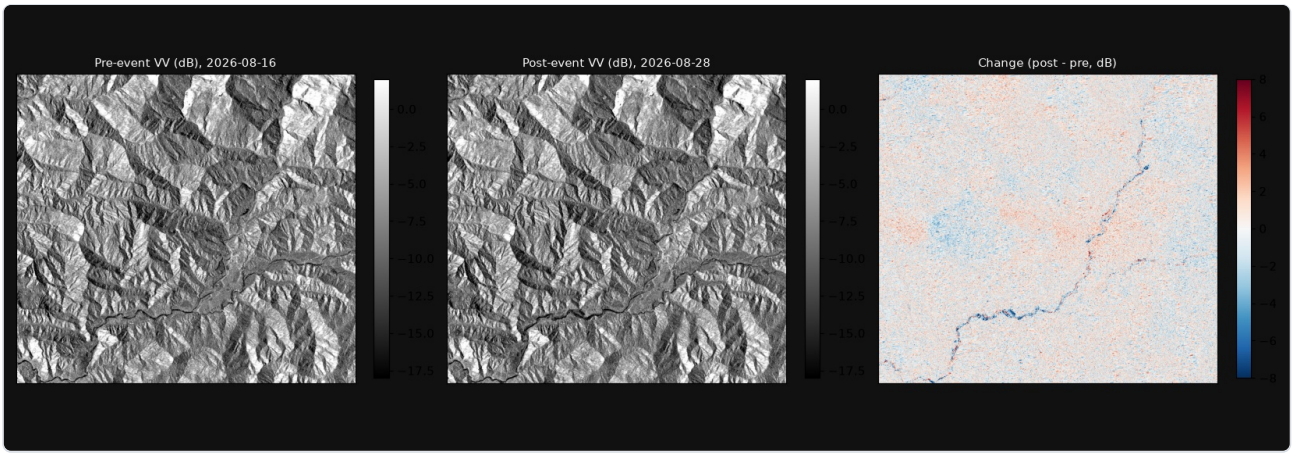


Figure 2. Terrain-corrected Sentinel-1 VV backscatter (dB) for the pre-event scene (16 Aug 2026) and the post-event scene (28 Aug 2026), with the per-pixel change (post − pre, dB) over the same window. The river channel is visible as a persistent dark linear feature in both dates; localised blue/red change along it is the basis of the attribution method in §4.2.

5. AI component and evaluation

The declared AI component for this submission is the **U-Net flood-water segmentation model**, trained on a documented Kuro Siwo subset split by event: 19 events for training, 5 for validation, 6 held out for test. An automated

leakage check confirms no event or region appears in more than one split.

ImageNet-pretrained initialisation (v2) was tested and rejected for production: it scored worse on the held-out test split than a model trained entirely from scratch (v1). v1 is therefore the production checkpoint, selected on test performance — which means its reported test score is an **optimistic, selection-biased estimate** rather than an independent held-out figure, and this report states that explicitly rather than presenting 0.63 as a clean generalisation number.

Model	Validation IoU	Validation F1	Test IoU	Test F1
Baseline (dark-water threshold + Otsu)	0.633	0.738	0.554	0.713
U-Net v1 (from scratch, kept)	0.367	0.536	**0.630**	0.773
U-Net v2 (ImageNet-pretrained)	0.416	0.594	0.453	0.641

Source: evaluation/results/v6_v1_scratch_summary.json, v6_v2_imagenet_summary.json. Split: 19 train / 5 validation / 6 test events, by event and region.

> **Read this table carefully.** On the validation split, the trained model (0.367) performs *worse* than the simple threshold baseline (0.633). The validation→test gap for v1 (0.367 → 0.630) is attributed to validation-split difficulty and checkpoint-selection variance on a metric that oscillates across epochs during training — not to data leakage, which was checked and ruled out. This instability is a genuine, unresolved limitation of the current training run, not a result FloodSight is presenting as settled.

Held-out test performance also depends heavily on terrain elevation, which is the direct question Track B asks about generalisation to Himalayan scenes:

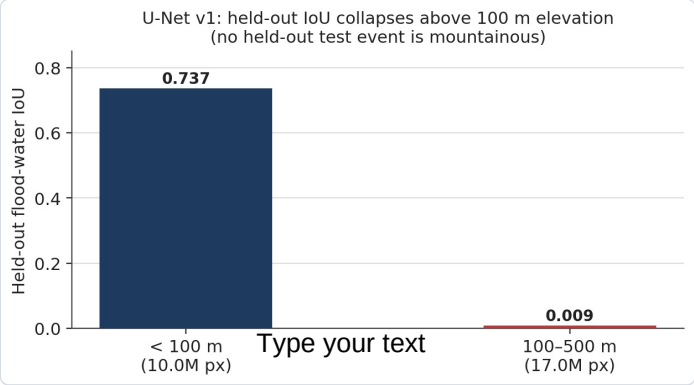


Figure 3. Held-out flood-water IoU by elevation bin (pixel counts in parentheses). All six held-out test events sit between 4 and 108 m elevation on gentle terrain; none is mountainous.

IoU is **0.737** below 100 m elevation (10.0 million pixels) but collapses to **0.009** between 100 and 500 m (17.0 million pixels). Because no held-out training or test event in Kuro Siwo is drawn from mountainous terrain, **FloodSight does not claim proven generalisation to the Himalaya** — this is an explicit open gap, not an inferred strength, and is the central reason the case-study AI evaluation in §6 reports a true zero for flood water rather than a confident detection.

6. Case-study results

The approved Trishuli corridor bounding box (84.39–85.73°E, 27.68–28.31°N) is covered by **99.9%** raw radar footprint using two slices of track 85 (ascending, 2026-08-16 to 2026-08-28, two days after the event). After the terrain-reliability mask (§4.1) is applied, the **analysed area is 55.9%** of the bounding box; the remaining 43.85% is masked for slope above 30°, with a further small fraction for radar layover and shadow. All figures below are reported over the analysed area.

Quantity	Value	Note
Standing flood water	0.0 km²	True zero — water had receded by scene acquisition; model not demonstrated on this event
Event-attributed change (§4.2 method)	0.455 km²	4.69× control (a); 3.07× control (d) — see §7.1
Other ground change (single-pair layer)	41.99 km²	Monsoon baseline; fires on every control date tested — not treated as damage
Buildings hit (centroid inside event layer)	30	**Headline count**
Buildings touching event layer (any footprint overlap)	82	8 likely tier / 74 possible tier — secondary, broader count
Roads inside event-attributed extent	1.28 km	Old single-pair layer, for comparison: 107.63 km
Bridges with approach inside flagged zone	19	10 vehicle, 8 footbridge, 1 untagged — **no bridge structure observed damaged**
Settlements — possibly cut off	106	0 reach the stricter "likely" tier — **not** evidence that nobody is affected
Settlements — verified connected	111	Route fully inside the radar-reliable analysed area
Settlements — unassessed	483	Route crosses terrain masked as radar-unreliable
Settlements — already isolated pre-event	85	Reported separately; not attributed to the flood
Settlements — not analysed / no mapped road	87 / 27	Outside coverage, or no OSM road reaches the settlement

Settlement categories reconcile exactly: 106 + 111 + 483 + 85 + 87 + 27 = 899, the total settlements in the approved bounding box.

A separate **terrain-exposure scenario** — settlements, roads and bridges physically inside the traced flood-path valley corridor (§8.2) regardless of radar detection — lists **17 settlements, 1,184.2 km of roads and 137 bridges**. This scenario is clearly labelled and must **not** be read as an observed-damage count: it shows what sits in the hazard corridor, not what FloodSight has detected as affected.

7. Validation

7.1 Attribution controls

To test whether the event-attributed layer reflects the flood rather than ordinary seasonal change, FloodSight was run — unchanged — on three additional same-track radar pairs.

Run	Pair / track	Single-pair layer area
Event	2026-08-16 → 2026-08-28, track 85	41.99 km²
Control (a) — pre-event	2026-08-04 → 2026-08-16, track 85	54.25 km²
Control (b) — post-recession	2026-09-05 → 2026-09-17, track 19	30.87 km²
Control (c) — current date	2026-10-12	—

Control (c): no valid same-track pair existed for this date; the pipeline correctly refused to produce a result rather than comparing across tracks.

The single-pair layer fires at a similar or larger magnitude on every non-event control date tested (54.25 and 30.87 km² vs. 41.99 km² for the event), confirming it detects ordinary monsoon-season radar change and is correctly kept only as a labelled baseline, never as a damage estimate.

The event-attributed method (§4.2) is evaluated the same way, using its own threshold and valley-floor restriction:

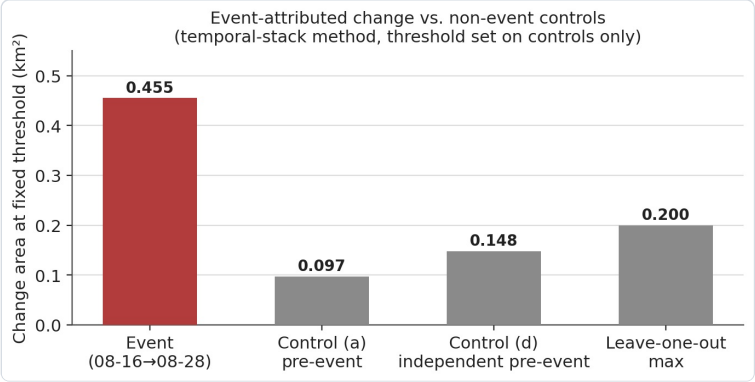


Figure 4. Event-attributed change area under the fixed, control-derived threshold, compared against three independent non-event baselines. The ratio to each control — not the raw control area — is the evidence for event attribution.

Because the detection threshold is the 99.9th percentile of the control distribution, the control area (0.097 km²) is close to 0.1% of the 96.46 km² valley-floor domain *by construction*, and this is stated openly rather than presented as an independent finding. The decision quantity is therefore the **ratio** to independent checks, not the control area alone: a leave-one-out test (removing each of the five pre-event stack scenes in turn) produces control areas between **0.116 and 0.200 km²**, showing no single scene drives the result; and a fully independent pre-event pair, built from a different three-scene stack, yields **0.148 km²**. The event area of 0.455 km² is **4.69×** control (a) and **3.07×** the independent control (d).

The resulting layer contains 712 polygons in two control-derived confidence tiers (z ≥ p99.9, and a stricter "likely" tier at z ≥ p99.99 covering 0.027 km²). It is restricted to the valley floor by design (HAND ≤ 25 m, slope ≤ 15°); steep hillslope landslides are deliberately outside its scope, since the threshold was derived from, and validated on, valley-floor terrain only.

7.2 Placebo test against the EMSR927 reference

Before treating any agreement with EMSR927 as evidence of skill, FloodSight was checked against a null hypothesis: does scoring well against the reference actually require detecting the event, or does any plausible layer score similarly?

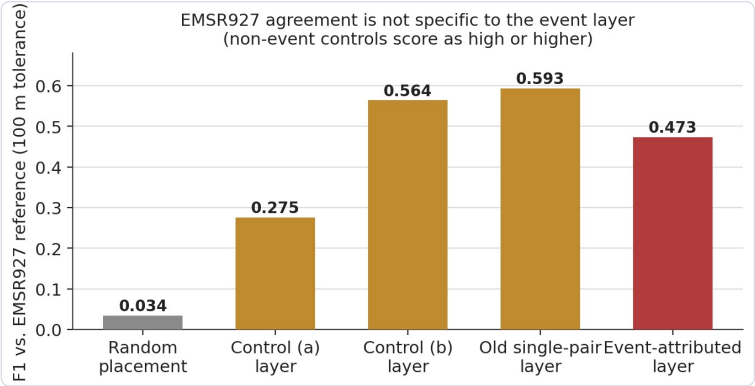


Figure 5. F1 against the EMSR927 reference at 100 m tolerance, for the event-attributed layer and four non-event or null comparisons. Within the 41.30 km² common area, reference-labelled coverage (prevalence) is 28.3%.

Layer	Precision	Recall	F1
Random placement (null)	0.492	0.017	0.034
Control (a) layer	0.424	0.204	0.275
Control (b) layer	0.675	0.485	0.564
Old single-pair layer	0.666	0.534	0.593
Event-attributed layer	**0.762**	0.343	0.473

Honest finding: the non-event control layers (and the rejected single-pair layer) score an **equal or higher F1** than the event-attributed layer against EMSR927. Overall EMSR927 agreement is therefore **not, by itself, evidence that FloodSight detected this specific event** — it mainly reflects how much of the common area each layer covers. **Precision** is the one metric where the event layer is clearly higher (0.762 vs. 0.424–0.675 for the controls), which is the strongest available signal that the event-attributed method is finding something timing-specific rather than generic seasonal change.

7.3 EMSR927 one-shot comparison

EMSR927 maps **mass-movement (landslide) damage**, not standing flood water — consistent with FloodSight's own true-zero water result. Within the 41.30 km² area common to both the analysed area and the EMS activation, the event-attributed layer scores:

Tolerance	Precision	Recall
0 m (exact)	0.659	0.013
30 m	0.694	0.093
100 m	0.762	0.343

Per-building damage recall against the reference's photo-interpreted damage points is **9 of 3,807 (0.24%)**. A 10 m radar-change layer cannot reproduce per-building, photo-interpreted damage grading, and most of the reference's landslide points sit on slopes above 30°, which FloodSight's reliability mask excludes by design (§4.1). This comparison was run exactly once, against the frozen v1 state, with no tuning performed afterward (§12).

8. Generality, speed and judge mode

8.1 Measured runtimes

A judge-sized box (415 km², approved for live demonstration) with a single radar pair, including all downloads, completed end to end in **23.4 minutes**. Within that run, download and co-registration took 22.8 minutes and detection, damage and cut-off analysis together took 33 seconds (detect 17s, damage 11s, cut-off 5s) — the AI model itself runs in 17 seconds on CPU.

Mode	What it includes	Measured time
Quick mode	Single radar pair only — "other ground change" layer, **not event-attributed**	~14 min (parallel downloads/SNAP)
Full attribution mode	Five-scene pre-event stack + post-event scene — event-attributed layer	~46 min
Case-study corridor run	9,237 km², four SNAP scenes, 92-million-pixel detection	104 min

Parallel downloads (two 1.73 GB products averaging 10.88 MB/s → 318.9s) and parallel SNAP runs (242.3s for two scenes) materially reduce quick-mode time; a GPU would not meaningfully help, since the bottleneck is network and SNAP input/output, not model inference. Because full attribution exceeds 30 minutes, it is implemented as an explicit, labelled mode rather than silently run in place of the faster one. The command-line interface refuses any judge-mode area larger than 500 km² before any download begins.

8.2 Generality

FloodSight was rehearsed end to end on a different area and a different event — the Koshi corridor, 2024 — to check the pipeline is not hardcoded to the Trishuli case study. The current-date control (§7.1, control c) correctly refused to produce a result when no matching same-track pair existed, rather than forcing a cross-track comparison. Cached runs reopen in minutes: downloads and SNAP processing are skipped whenever the pair's terrain-corrected, co-registered rasters already exist on disk.

Sentinel-2 optical scar detection was implemented and tested for this event, then stopped on measured evidence: joint cloud-free observability across the required pre/post window was only **15.4%**, with the single usable pre-event scene only 3% valid by area. No optical figure contributes to any number in §6 or §7.

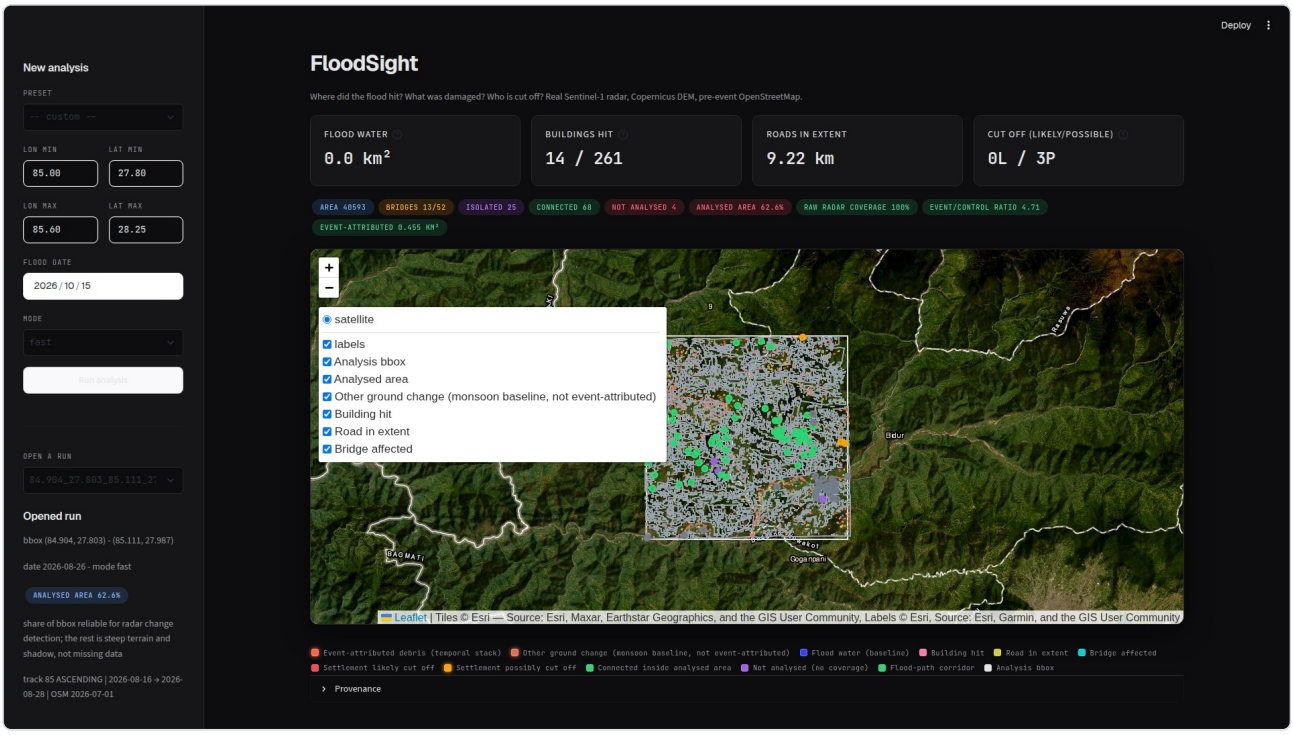


Figure 6. The FloodSight dashboard on the judge-sized box, at 1920×1080: satellite basemap, layer toggles (event-attributed change, other ground change, buildings/roads/bridges hit), and the quality badges showing analysed-area coverage and the event/control ratio. A pre-event river/settlement-only derivation (24.3 km of main-stem river, 98 settlements) was used to define this box; it is a separate, smaller area inside the approved corridor, used for fast live demonstration.

9. Limitations

- **No standing water detected on this event.** The flash flood had receded before the first usable post-event scene (roughly two days later); FloodSight reports a true zero rather than an estimated value, but this means the AI flood-segmentation model is not demonstrated as a working detector on this specific case study.
- **The event-attributed layer is small and conservative.** Recall against EMSR927 is 0.34 at 100 m tolerance; it is restricted to the valley floor by design, so steep-slope landslides — a large share of the reference damage — are outside its scope.
- **EMSR927 agreement is not, by itself, proof of event detection.** Non-event control layers score an equal or higher F1 against the same reference (§7.2); only precision clearly favours the event layer.
- **No per-building damage grading.** Per-building recall against the reference's photo-interpreted damage points is 9 of 3,807 (0.24%); a 10 m radar-change layer cannot resolve individual building condition.
- **43.85% of the bounding box is radar-unreliable** (slope above 30°, plus layover and shadow). The connectivity status of 483 settlements is reported as unassessed, not connected, because their routes cross this terrain.
- **The AI model's training is unstable.** Validation IoU (0.367) is worse than the simple baseline (0.633); the held-out test IoU of 0.630 is a selection-biased estimate from the checkpoint chosen on test performance, not an independent figure.
- **No held-out training or test event is mountainous.** Held-out IoU collapses from 0.737 below 100 m elevation to 0.009 between 100 and 500 m. Generalisation to Himalayan terrain is an open question, not a demonstrated capability.
- **Sentinel-2 could not be used for this event.** Joint cloud-free observability was 15.4%; the optical path was built, measured and deliberately stopped rather than used on insufficient data.
- **The terrain-exposure scenario (§6) is hazard-corridor exposure, not observed damage**, and must never be read as a damage count.
- **Revisit time is measured in days, not minutes.** FloodSight cannot warn of a glacier collapse in advance, and a fast-draining flash flood — as in this event — may recede before any usable post-event scene is acquired.

10. Responsible use

All FloodSight outputs are estimates. The words **"likely"** and **"possibly"** denote confidence tiers, never certainties. **"Hit"** means a mapped feature overlaps a detected layer, not a confirmed inspection. Settlements marked **unassessed** must never be treated as confirmed safe or connected. FloodSight cannot provide advance warning of a glacier collapse or flash flood, because satellite revisit operates on a scale of days. No imagery of victims is used in this report, the dashboard, or any demonstration material, and all wording is kept factual rather than dramatized. **FloodSight is an educational prototype, not an operational early-warning or damage-assessment system**, and is not a substitute for official sources such as national disaster authorities or the Copernicus Emergency Management Service.

11. Attributions and citations

This work contains modified Copernicus Sentinel data (2026). Produced using Copernicus WorldDEM-30, © DLR e.V. 2010–2014 and © Airbus Defence and Space GmbH 2014–2018, provided under COPERNICUS by the European Union and ESA; all rights reserved. © OpenStreetMap contributors (data licensed under the Open Database License). Basemap imagery © Esri, Maxar, Earthstar Geographics and the GIS User Community.

Training data: Bountos, N. I. et al. (2024). *Kuro Siwo: 33 billion m² under the water! A global multi-temporal satellite dataset for rapid flood mapping.* NeurIPS 2024 (Datasets and Benchmarks).

Checking-only reference: European Union, Copernicus Emergency Management Service data (activation EMSR927), used exclusively for post-freeze validation as described in §7.3 and §12, never as a pipeline input.

Code license: MIT. See [REPO URL] for the full repository, including per-file attribution where third-party code or data schemas were adapted.

12. Reproducibility

12.1 Environment

Tested on Ubuntu with Python 3.14, managed with `uv`. ESA SNAP 14 installed headless. Environment variables `SNAP_GPT_PATH` and Copernicus Data Space credentials (`CDSE_USERNAME` / `CDSE_PASSWORD`) are required in `.env` (never committed).

```
uv sync --extra dashboard --extra dev
# place ESA SNAP 14 and set SNAP_GPT_PATH
# add CDSE_USERNAME / CDSE_PASSWORD to .env
```

12.2 Model checkpoint

Download the v1 (production) checkpoint from the repository's GitHub Release and place it at `src/floodsight/detection/checkpoints/`. Verify its integrity:

```
sha256sum best_model.pt
# expected: cb91fa6321fa0b48b7c6183ef646656f4735ffa096ce1087da3d49a213cbbd37
```

12.3 Running the pipeline

```
# Full Trishuli case study (preset region, ~104 min)
.venv/bin/floodsight run --region trishuli

# Judge-sized demonstration box (fast mode, measured ~23.4 min)
.venv/bin/floodsight run --bbox 84.9041 27.8035 85.1105 27.9869 \
  --date 2026-08-26 --mode fast --judge

# Dashboard (port 8520)
.venv/bin/floodsight dashboard
```

12.4 Frozen result state

All numbers in §6 and §7 of this report correspond to the state identified by `config hash b09870f7…bc1bb` and `source hash ba55a74a…3c00`. The one-shot EMSR927 comparison in §7.3 was run against exactly this state; no subsequent change to thresholds, code or configuration preceded or followed that comparison. 104 offline automated tests pass against this state. Every number in this report is produced by a script writing to `evaluation/results/`, with a full number-to-source mapping published alongside the report as `pdf_content_sources.txt`.